

Linux Commands

System Management

System Info Commands:

Command	Description	Variants	Example	Output / Use Case
<code>uname -a</code>	Shows complete system information (kernel, architecture, OS).	<code>-r</code> (kernel release), <code>-s</code> (kernel name), <code>-m</code> (machine type)	<code>uname -a</code>	Displays kernel version, OS type, and architecture — useful for system troubleshooting and compatibility checks.
<code>hostname</code>	Prints the system's hostname (computer name).	<code>-I</code> (show IP), <code>--fqdn</code> (full hostname)	<code>hostname -I</code>	Shows the system's hostname or IP — useful in networking and server setups.
<code>top</code>	Real-time process viewer showing CPU and memory usage.	<code>-o</code> (sort), inside <code>top</code> : <code>P</code> (CPU sort), <code>M</code> (memory sort)	<code>top</code>	Monitors system load and running processes — helps diagnose system lag.
<code>htop</code>	An improved, colorful, interactive version of <code>top</code> .	F10 (exit), F6 (sort), F9 (kill process)	<code>htop</code>	Provides better UI for monitoring processes — widely used in DevOps.
<code>df -h</code>	Displays disk usage of all mounted filesystems.	<code>-h</code> (human readable), <code>-T</code> (filesystem type)	<code>df -h</code>	Helps identify disk space usage — essential to prevent “disk full” issues.
<code>du -sh</code>	Shows total size of a directory.	<code>-h</code> (human readable), <code>-s</code> (summary), <code>-a</code> (all files)	<code>du -sh /var/log</code>	Identifies which directories consume the most space — helps fix storage issues.
<code>free -m</code>	Shows RAM usage (used, free, available).	<code>-h</code> (human readable), <code>-g</code> (GB)	<code>free -m</code>	Helps diagnose memory-related performance issues.

Package Management Commands:

Command	Description	Variants	Example	Output / Use Case
<code>apt</code>	Package manager for Debian/Ubuntu systems. Used to install, update,	<code>apt update</code> (refresh list), <code>apt upgrade</code> , <code>apt</code>	<code>sudo apt install nginx</code>	Installs Nginx on Ubuntu; very commonly used in servers and

	remove packages.	install pkg, apt remove pkg		DevOps environments.
yum	Package manager for older RHEL/CentOS systems.	yum install pkg, yum update, yum remove pkg	sudo yum install vim	Used to manage packages on CentOS 7/RHEL 7 systems.
dnf	Modern package manager for Fedora, RHEL 8+, CentOS 8+ ; replacement for yum.	dnf install pkg, dnf update, dnf remove pkg	sudo dnf install curl	Default package manager in new RedHat-based distros.
snap	Universal package manager that installs sandboxed apps across many Linux distros.	snap install pkg, snap remove pkg, snap list	sudo snap install core	Installs applications with automatic updates — widely used for tools like VS Code, Docker, etc.
dpkg	Low-level Debian package manager used to install .deb files manually.	-i (install), -r (remove), -l (list installed)	sudo dpkg -i package.deb	Installs .deb files downloaded manually — useful when package is not in apt repository.

Server Management Commands:

Command	Description	Variants	Example	Output / Use Case
systemctl	Controls system services (start, stop, enable, disable, check status). Works on systemd-based Linux .	systemctl start svc, systemctl stop svc, systemctl restart svc, systemctl enable svc, systemctl status svc	sudo systemctl start apache2	Starts Apache service; used in almost every Linux server to manage services.
journalctl	Shows logs for systemd services. Very useful for debugging errors.	-u svc (service logs), -f (live logs), --since (time filter)	sudo journalctl -u ssh -f	Streams real-time SSH logs — helps trace issues or login attempts.
service	Older wrapper for managing services (works on SysVinit and systemd).	service svc start, service svc stop, service svc status	sudo service apache2 status	Used for backward compatibility; works on older distros or scripts.

Disk Management Commands:

Command	Description	Variants	Example	Output / Use Case
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lsblk	Lists all block devices (disks, USBs, partitions).	-f (show filesystem), -a (all devices)	lsblk -f	Helps you see available disks, partitions, mount points — first step before partitioning.
fdisk	Used to create, delete, modify disk partitions (MBR disks).	fdisk -l (list disks), inside menu: n new, d delete, w write	sudo fdisk /dev/sdb	Creates partitions on a disk — essential for manual storage setup.
mount	Mounts a filesystem or partition to a directory.	-t type, -o options	sudo mount /dev/sdb1 /mnt	Allows you to access a disk/partition by attaching it to a directory.
umount	Unmounts an attached filesystem.	(none)	sudo umount /mnt	Safely detaches a mounted partition — needed before formatting or ejecting.
df	Shows disk usage of mounted filesystems.	-h (human readable), -T (show filesystem type)	df -h	Used to check available and used disk space — prevents storage issues.

Process Management Commands:

Command	Description	Variants	Example	Output / Use Case
ps	Shows running processes for the current user or system.	aux (all processes), ef (tree view)	ps aux	Lists all processes with PID — first step before killing a stuck process.
kill	Sends a signal to terminate a process using PID.	-9 (force kill), -15 (safe kill)	kill -9 1234	Kills a specific process that is frozen or misbehaving.
killall	Kills all processes by name instead of PID.	(none)	killall firefox	Ends all Firefox processes — helpful when multiple instances hang.
nice	Starts a process with a specific priority.	-n value	nice -n 10 bash script.sh	Lower/higher CPU priority for tasks — useful for performance management.
renice	Changes priority of an already running process.	High priority: -20, Low priority: 19	sudo renice -10 1234	Adjusts priority of heavy processes to avoid system lag.
jobs	Shows background jobs in the current terminal.	(none)	jobs	Useful for managing suspended/background tasks.
bg	Resumes a paused job in the	(none)	bg %1	Moves job 1 to background so terminal is free.

	background.			
fg	Brings a background job to the foreground.	(none)	fg %1	Useful when you need to interact with a job again.

File Compression & Archiving Commands:

Command	Description	Variants	Example	Output / Use Case
zip	Compresses files/directories into a .zip archive.	-r (recursive), -9 (max compression), -q (quiet mode)	zip -r backup.zip /home/abdullah	Creates a compressed ZIP file — sharing & backup ke liye useful.
unzip	Extracts .zip files.	-l (list), -d (extract to directory)	unzip backup.zip -d extracted/	Unpacks ZIP archives — downloaded files or backups extract karne ke liye.
tar	Creates & extracts tar archives (.tar, .tar.gz, .tgz).	-c (create), -x (extract), -v (verbose), -f (file), -z (gzip), -j (bzip2)	tar -czvf logs.tar.gz /var/log	Creates compressed backups; DevOps/servers me most used.
gzip	Compresses a single file into .gz.	-d (decompress), -k (keep original), -v (details)	gzip file.txt	Creates file.txt.gz — logs compress karne me common.
gunzip	Decompresses .gz files.	(none)	gunzip file.txt.gz	Restores original file — for reading or analyzing logs.